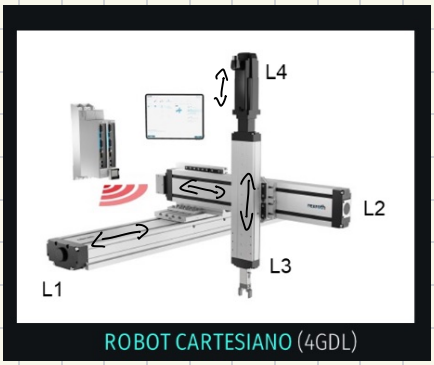

ROBOT CARTESIANO (4GDL)

① $A_1 - A_2$

$\boxed{\text{Rotacion}}$ de 90° positiva en X

$$\begin{bmatrix} \cos(\theta_1) & -\text{sen}(\theta_1) & 0 \\ \text{sen}(\theta_1) & \cos(\theta_1) & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix} = \begin{bmatrix} \cos(\theta_1) & 0 & \sin(\theta_1) \\ \sin(\theta_1) & 0 & -\cos(\theta_1) \\ 0 & 1 & 0 \end{bmatrix}$$

$\boxed{\text{Traslacion}}$

$$\begin{bmatrix} 0 \\ 0 \\ -L_1(t) \end{bmatrix}$$

② $A_2 \to A_3$

$\boxed{\text{Rotacion}}$ de 90° positiva en Y

$$\begin{bmatrix} \cos(\theta_1) & -\text{sen}(\theta_1) & 0 \\ \text{sen}(\theta_1) & \cos(\theta_1) & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ -1 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & -\sin(\theta_1) & \cos(\theta_1) \\ 0 & \cos(\theta_1) & \sin(\theta_1) \\ -1 & 0 & 0 \end{bmatrix}$$

$\boxed{\text{Traslacion}}$

$$\begin{bmatrix} 0 \\ 0 \\ -L_2(t) \end{bmatrix}$$

③ $A_3 \to A_4$

$\boxed{\text{Rotacion}}$ No hay rotacion

$$\begin{bmatrix} \cos(\theta_1) & -\text{sen}(\theta_1) & 0 \\ \text{sen}(\theta_1) & \cos(\theta_1) & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$\boxed{\text{Traslacion}}$

$$\begin{bmatrix} 0 \\ 0 \\ L_3(t) \end{bmatrix}$$